

# Hormonal growth promoting agents in food producing animals.

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In contrast to the use of hormonal doping agents in sports to enhance the performance of athletes, in the livestock industry hormonal growth promoters ("anabolics") are used to increase the production of muscle meat. This leads to international disputes about the safety of meat originating from animals treated with such anabolics. As a consequence of the total ban in the EU of all hormonal active growth promoters ("hormones") in livestock production, in contrast to their legal use [e.g. of five such hormones (17 $\beta$ -estradiol, testosterone, progesterone, trenbolone and zeranol) as small solid ear implants and two hormones as feed additives for feedlot heifers (melengestrol acetate) and for swine (ractopamine) in the USA], the regulatory controls also differ sharply between the EU and the USA. In the EU the treatment of slaughter animals is the regulatory offence that has to be controlled in inspection programs. In the USA testing for compliance of a regulatory maximum residue level in the edible product (muscle, fat, liver or kidney) is the purpose of the inspection program (if any). The EU inspection programs focus on sample materials that are more suitable for testing for banned substances, especially if the animals are still on the farm, such as urine and feces or hair. In the case of slaughtered animals, the more favored sample materials are bile, blood, eyes and sometimes liver. Only in rare occasions is muscle meat sampled. This happens only in the case of import controls or in monitoring programs of meat sampled in butcher shops or supermarkets. As a result, data on hormone concentrations in muscle meat samples from the EU market are very rare and are obtained in most cases from small programs on an ad hoc basis. EU data for natural hormones in meat are even rarer because of the absence of "legal natural levels" for these hormones in compliance testing. With the exception of samples from the application sites - in the EU the site of injection of liquid hormone preparations or the site of application of "pour on" preparations - the hormone concentrations observed in meat samples of illegally treated animals are typically in the range of a few micrograms per kilogram (ppb) down to a few tenths of a microgram per kilogram. In the EU dozens of illegal hormones are used and the number of active compounds is still expanding. Besides estrogenic, androgenic and progestagenic compounds also thyreostatic, corticosteroidal and beta-adrenergic compounds are used alone or in "smart" combinations. An overview is given of the compounds identified on the EU black market. An estimate is also given of the probability of consumption in the EU of "highly" contaminated meat from the application sites in cattle. Finally some data are presented on the concentration of estradiol in bovine meat from animals treated and not treated with hormone implants. These data are compared with the recent findings for estradiol concentrations in hen's eggs. From this comparison, the preliminary conclusion is that hen's eggs are the major source of 17 $\alpha$ - and 17 $\beta$ -estradiol in the consumer's daily "normal" diet.